

Individual Homework 1 — Applied ML for Business

Due: TBD | Individual | 95 points

Submission: Upload [studentid].md on Moodle (DLE). Answers must be typed in markdown format. Use the question numbering exactly as it appears here; matching is essential for grading.

On graphs: Where a question asks you to generate a plot, write your observations and interpretation in your answer. No images or PNGs are required in your submission file.

Individual submission. This assignment is designed to help you recall and consolidate topics discussed in class. Spend adequate time with each question; the goal is understanding, not completion speed.

Q1 — Hospital PCA [4 pts]

A hospital tracks 8 features for ICU patients on admission: age, blood pressure, creatinine, albumin, BMI, cholesterol, heart rate, and prior admissions. PCA is run on standardised features. PC1 explains 49% of variance. All 8 features have positive loadings on PC1.

- (a) What is PC1 capturing about patients — and why do all-positive loadings make clinical sense?
 - (b) All PC1 loadings are positive. Can a patient still have a negative PC1 score? Explain.
 - (c) Patient A: PC1 score +2.4. Patient B: PC1 score -1.9. The admissions team proposes using PC1 score to prioritise the 10 available ICU beds. A senior clinician objects: “PC1 tells you about systemic burden. ICU decisions are about acute risk, these are not the same thing.” Read up briefly on ICU triage and describe a scenario involving two patients, with brief clinical particulars for each, where admitting the higher PC1 patient over the lower PC1 patient leads to a bad outcome.
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Q2 — Recipe Nutrition [3 pts]

PCA is run on 500 recipes using 7 raw (unstandardised) features: calories, protein, fat, carbs, fiber, sugar, sodium. The PC1 loading on calories is 0.97; all other loadings are below 0.15.

The nutritionist concludes: “Our analysis shows calories is the only dimension that matters in nutrition.” Is she right? What would change her conclusion — and what does that tell you about a decision made before running PCA?

Q3 — Index Fund Rebalancing [8 pts]

You are a data analyst at a brokerage. Each quarter the NIFTY 50 reconstitutes — stocks enter or exit based on market capitalisation. When a stock enters, every fund tracking the index must hold it in exact proportion to its assigned weight. The gap between what the fund holds and what the index requires is tracking error — fund managers are penalised for it. Your team has daily returns for 200 NSE-listed stocks over 3 years (~750 trading days). Plaksha_Fab, currently priced at Rs.500, is flagged as a likely addition with weight 1.5%. When the addition is announced, traders buy in anticipation — pushing the price to Rs.620 before reconstitution day. The fund buys at Rs.620.

- (a) [1 pt] Describe the data matrix: rows, columns, and what one entry represents.
 - (b) [1 pt] Alice sets rows = stocks, Bob sets rows = days; which is correct for understanding co-movement and what does one row represent?
 - (c) [1 pt] After standardising returns, the team computes the covariance matrix; what is it equivalent to and what are its dimensions?
 - (d) [1 pt] PC1 has positive loadings on almost all 200 stocks; what does this tell you about how stocks move?
 - (e) [2 pts] Reliance (weight 8.5%) has PC1 loading 0.28; a smaller stock (weight 0.4%) has loading 0.06; why does Reliance load more heavily on PC1? (Recall the class discussion on the market factor.)
 - (f) [2 pts] Day tau: PC1 score +3.1. Day sigma: PC1 score -2.8. What happened to most stocks on each day? Give a real-world event that could explain each.
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Q4 — CLV Derivation [10 pts]

A customer generates margin m per period, stays with probability r each period, and future cash flows are discounted at rate d .

- (a) [3 pts] Write the expected discounted profit for periods 0, 1, 2, 3.
 - (b) [2 pts] Identify the first term a and common ratio q of the geometric series.
 - (c) [5 pts] Sum the series to derive $CLV = m(1+d)/(1+d-r)$; show your work.
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Q5 — Retention Sensitivity [7 pts]

Given $m = \text{Rs.}100$, $d = 0.10$:

- (a) [2 pts] Compute CLV for $r = 0.80, 0.85, 0.90, 0.95$.

- (b) [2 pts] Plot CLV vs r from 0.50 to 0.99.
- (c) [3 pts] Compute the CLV gain from $r = 0.80$ to 0.85 and from $r = 0.90$ to 0.95; report both and explain why they differ.
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Q6 — Manager’s Fallacy [5 pts]

The manager says: “We increased retention from 85% to 90% — that’s 5 percentage points, so CLV improved by 5%.” Show why this is wrong using your calculations.

Q7 — Why Two Models [3 pts]

Your team is building a model to predict how much a returning customer will spend next month. Explain why you need two models — a return classifier and a spend regressor — instead of one model predicting spend directly.

Q8 — AUC vs Calibration [8 pts]

A streaming platform’s analytics team presents to the VP of Marketing: “We built a churn prediction model on our 3.2 million subscribers. $AUC = 0.88$ on the held-out test set. We scored all subscribers, identified the 15,000 highest-risk customers, and used their predicted churn probabilities to compute CLV. This group averages CLV of Rs.3,200 — roughly half the platform average. We recommend a targeted retention offer of Rs.800 per customer, well within their CLV.” The VP approves the campaign citing the AUC.

- (a) [2 pts] What does $AUC = 0.88$ tell you about this model — and what does it not tell you?
- (b) [2 pts] The model outputs $P(\text{churn in the next period})$; the team plugs this into the CLV formula as the retention rate r . Are these the same quantity? What is the problem?
- (c) [2 pts] The model’s predicted churn probabilities for the 15,000 highest-risk customers are compressed between 0.52 and 0.61, when their true churn rates are 0.75–0.90; are the CLV estimates overstated or understated, and what does this imply for the Rs.800 voucher decision?
- (d) [2 pts] AUC measures how well the model ranks customers by churn risk. Why is good ranking not sufficient here? What property of the model’s outputs does the CLV computation require instead?
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Q9 — Supervised vs Unsupervised [10 pts]

A streaming education platform has 180,000 subscribers. For each user: x_i in $\mathbb{R}^8$ — sessions, avg_duration, courses_started, courses_completed, completion_rate, recency, streak, quiz_share. 40,000 subscriptions have expired; for these the platform knows y_i — sessions in the final 30 days before expiry. The remaining 140,000 are active; y_i unknown.

Analyst A runs K-means ($k=4$) on x_i for all 180,000 users. Analyst B trains simple regression on the 40,000 expired users, generates $\hat{y}_i$ for all 180,000, then runs K-means ($k=4$) on $\hat{y}_i$ alone.

- (a) [2 pts] Write the objective that Analyst A's K-means minimises; write the objective that Analyst B's regression minimises; state what variable appears in one but not the other.
 - (b) [3 pts] Analyst A's clusters group users by similarity in x_i ; Analyst B's by similarity in $\hat{y}_i$; what is each clustering grouping users by, and what can each tell you that the other cannot?
 - (c) [3 pts] The retention manager has budget for 5,000 vouchers and wants to reach users most likely to go inactive; whose output should she use, and why does the other approach structurally fail her question?
 - (d) [2 pts] Analyst B trained on expired subscribers and predicts on active ones; what must be true for this to produce valid predictions?
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Q10 — Spotify Audio Features [24 pts]

Load `spotify_songs.csv` from the course GitHub. The dataset contains audio features for tracks across 6 genres.

- (a) [1 pt] Report the number of rows and columns in the full dataset.
- (b) [1 pt] List the numerical audio features; state which columns you would exclude from PCA and why.
- (c) [1 pt] Sample 250 tracks per genre using `random_state=42`; report your final dataset shape.
- (d) [2 pts] Standardise your features using `StandardScaler` from `sklearn.preprocessing`; explain why standardisation is necessary here and what goes wrong if you skip it.
- (e) [1 pt] Before computing anything: predict one pair of features you expect to be strongly positively correlated and one pair strongly negatively correlated; state your reasoning.
- (f) [2 pts] Compute the correlation matrix; report the strongest positive and strongest negative pair and state whether your predictions held.
- (g) [2 pts] Run PCA using `PCA` from `sklearn.decomposition`; report the variance explained

by each of the first 5 components and state how many you would retain and why.

(h) [2 pts] Report the PC1 loading for each feature; describe what PC1 captures and name the two features that drive your interpretation.

(i) [2 pts] Report the PC2 loading for each feature; describe what PC2 captures.

(j) [1 pt] Pick 3 songs you personally listen to; search for them by `track_name` and report the track name, artist, and genre for each. If a song is not in the dataset, pick the closest substitute.

(k) [1 pt] Report the PC1 and PC2 score for each of your 3 songs.

(l) [3 pts] Plot all 1,500 sampled tracks as a scatter in PC1–PC2 space, coloured by genre; mark your 3 songs as distinct points with names labelled and include the plot in your submission.

(m) [2 pts] Do your 3 songs sit where you would expect given your labels for PC1 and PC2? Explain for 2 of your songs.

(n) [2 pts] Compute the mean PC1 and PC2 score by genre; report which genre sits highest on PC1 and lowest on PC2, and state whether this matches what PC1 and PC2 represent.

(o) [1 pt] Compute the correlation between `track_popularity` and PC1, and between `track_popularity` and PC2; report both values. What does this tell you about the relationship between a song’s audio characteristics and its popularity?

(p) [2 pts] Spotify wants to build a “Focus” playlist for deep work; using your PC1 and PC2 labels, describe which region of the PC space you would target and why.

Q11 — RISK vs LIFT [13 pts]

In class we studied a telecom company’s retention programme. The firm predicted each customer’s churn probability, ranked them by RISK, and assigned retention offers to the highest-risk customers — a strategy that largely failed.

The firm has 10,000 customers. The retention offer costs Rs.200 per customer and recovers Rs.1,200 in revenue if the customer stays.

(a) [3 pts] Describe RISK and LIFT as discussed in class; explain why high RISK does not imply high LIFT.

(b) [4 pts] Customer A has 90% predicted churn probability and 5% LIFT. Customer B has 40% predicted churn probability and 35% LIFT. Compute the expected profit from offering the retention deal to each; who do you target?

(c) [1 pt] The data scientist runs a randomised pilot: 1,000 customers split equally into treatment (offered the deal) and control. She then trains a supervised model on the pilot data. What is her training label? How does it differ from the label in a standard churn model?

- (d) [1 pt] What would she ideally observe for each customer — and why is that impossible?
- (e) [2 pts] What goes wrong if she skips the pilot and estimates LIFT from historical data where only high-risk customers received the offer?
- (f) [2 pts] What is a “sleeping dog” in this context? Why is targeting a sleeping dog with a retention offer worse than ignoring them?
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